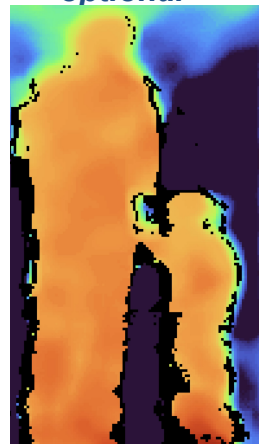


RGB crop + bbox



Metric depth
optional



Ground plane
optional

RGB encoder
DINOv3 ViT-H+/16

LoRA

4 8 16 24

gated depth fusion

DINO
ViT-S/16

patch-embed
+ proj

depth encoder

32×32
depth tokens

+ global
depth token

feat. + mask + pos.

Linear 11 → 1024

11-D geometry

❄️ Frozen 🔥 Trained 🟡 RGB 🔵 Depth 🟣 Ground 🟢 SAM 3D Body - - - optional input

Body decoder

LoRA

self + cross

Depth

cross-attention

Hand decoder

x6

Ground cond.
MHR token
cam token
kp prompts
*3-D kps
*2-D kps
*hand pos.
*learnable

MHR head

MHR body
model

Camera head

Depth-at-joint refiner



Δ pose

corrected pose

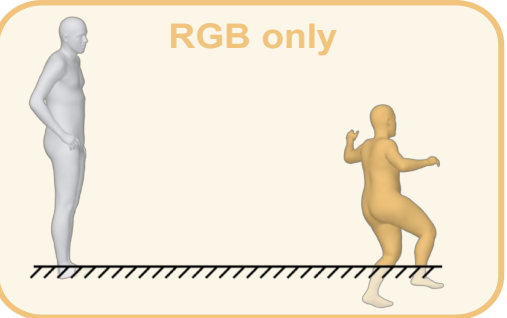
Ground Contact
Correction

Needs camera
height/orientation

same floor plane · k = 1 if tests fail

Depth Surface
Correction

RGB only



RGB-Ground

RGB-D